

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
END TERM EXAMINATION-MAY 2025

Course: B.Tech
 Subject Name: POWER SYSTEM

Semester: IV
 Subject Code: EE-221
 Maximum Marks: 50

Time Duration: 3 Hours

Note: Assume necessary data if required.

S.No	Question	Marks	CO							
1	A) Draw and briefly explain the various parts of a typical 33 kV substation.	3	CO-5							
	B) Explain the different types of the substations.	4	CO-5							
	C) Do we use AC or DC to charge a battery?	1	CO-5							
	D) List out the equipment used in a substation.	2	CO-5							
2	A) An 11 kV, 2.5 km, 3-phase, 3-wire belted cable gave the following results in capacitance testing. i. Capacitance between two conductors joint to sheath and the third conductor = 1.5 μ F. ii. Capacitance among all three conductors joint and sheath = 1.8 μ F. Then find (a) the capacitance of each core to neutral. (b) Capacitance between any two cores. (c) Charging current.	6	CO-4							
	B) A DC distributor cable is 1,000 m long and is loaded as under. <table border="1" style="margin: 10px auto; width: 60%;"> <tr> <td>Distance from the feeding point "A"</td><td>250 m</td><td>750 m</td><td>1,000 m</td></tr> <tr> <td>Load</td><td>100 A</td><td>200 A</td><td>300 A</td></tr> </table> The resistance of each conductor is 0.025 Ω per 1,000 m. Find the voltage at each load point, if the voltage at the feeding point "A" is maintained at 250 V.	Distance from the feeding point "A"	250 m	750 m	1,000 m	Load	100 A	200 A	300 A	4
Distance from the feeding point "A"	250 m	750 m	1,000 m							
Load	100 A	200 A	300 A							
3	A) A three phase, 50 Hz transmission line has resistance, inductance and capacitance per phase of 10 Ω , 0.1 H and 0.9 μ F and delivers a load of 35 MW at 132 kV and 0.8 pf lag. Determine the efficiency and regulation of the line using nominal T method.	6	CO-3							
	B) A three phase, 220 kV, 50 Hz transmission line consists of 2 cm radius conductor spaced 2.5 m apart in equilateral triangular formulation. If the temperature is 20° C and atmospheric pressure 75 cm, $m_o = 0.80$. Determine the corona loss per km of line.	4	CO-3							
4	A) The line supports of an overhead line are 180 m apart and are at the same level. The line conductor is of hard drawn copper having a diameter of 1.27 cm. Calculate the maximum sag (a) in still air (b) in a wind pressure of 33.7 kg per m ² of the projected area of	5	CO-2							

the ice-covered line. The radial thickness of the ice is 1.25 cm.

Assume that the maximum stress in the line is not to increase one-fifth of the ultimate strength 4,220 kg per cm². The weight of conductor per meter run is 1.13 kg.

- B) For the string of 3 insulators shown in Fig. 1. Calculate,
 (a) the voltage across each insulator as a percentage of line voltage to earth
 (b) the string efficiency.

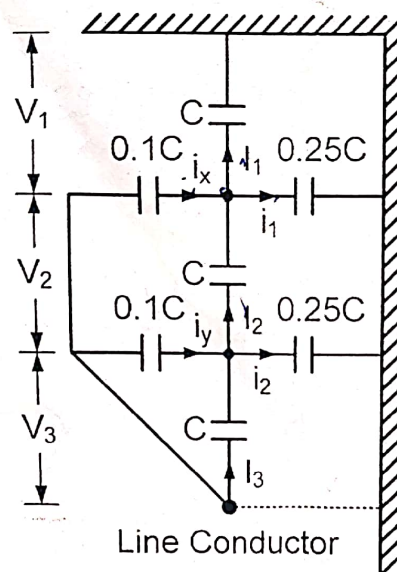


Fig. 1

- C) Explain different types of insulators used in power system.

5

- A) Explain surge impedance loading with respect to an overhead transmission line. Show that

$$Z_c = \sqrt{Z_{oc} Z_{sc}}$$

- B) Calculate the loop inductance per km of a single-phase transmission line comprising of two parallel conductors 1 m apart and 1.25 cm in dia. Also calculate the reactance of the transmission line. Frequency is 50 Hz.

- C) If a concentrically stranded ACSR cable with total annular space filled with strands of uniform diameter of 1.5 cm has 4 layers. Calculate the total number of strands and overall diameter of the stranded conductor.

2

CO-2

5

CO-3

3

CO-1

2

CO-1